

Lustjinn

Python as a backend, for a .NET developer

A course that grows with the code. This edition was built on 1 October 2026.

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CHAPTER 0

Orientation

You already know how to build a backend. You know what a request pipeline is, what dependency injection buys you, why migrations exist, how `async/await` frees a thread. None of that changes. What changes is the vocabulary, the tooling, and a handful of assumptions so deep in .NET that you no longer notice them — and those are the ones that bite.

This course follows **Lustjinn** as it is built: a web app for roleplay with a memory that does not forget, the Python successor to `airp`. Each chapter is one step of the roadmap. Claude writes the code; the chapter explains it — what each new piece is, why it is there, and what it would be in .NET. When a chapter and the code disagree, the chapter is out of date: that is a bug.

0.1 How to read this course

Every chapter has the same shape: what we are building in this step, the code that was written, the concepts it introduced, and what changed in the repository. Four kinds of box recur:

THE .NET LENS

Maps a Python idea onto the .NET one you already know — and says where the analogy breaks.

WORTH KNOWING

Background that is useful but not needed to follow the step.

WATCH OUT

A trap. Usually one that would compile — or rather, *run* — without complaint.

TRY IT

A small exercise. Optional, but the fastest way to make a concept stick.

0.2 The map

Most of the Python toolchain has a direct counterpart in what you use every day. Keep this table nearby; later chapters assume it.

You know (.NET)	Lustjinn uses (Python)	What it is
<code>dotnet CLI</code>	uv	Creates projects, installs packages, runs things
<code>nuget.org</code>	PyPI	The public package registry
<code>.csproj</code>	<code>pyproject.toml</code>	The project file: name, dependencies, tool settings
<code>packages.lock.json</code>	<code>uv.lock</code>	Exact resolved versions, committed
<code>global.json</code>	<code>.python-version</code>	Pins the runtime version
Roslyn analyzers, <code>dotnet format</code>	ruff	Linters and formatters in one
The compiler + nullable reference types	pyright (strict)	The type checker
xUnit	pytest	Tests
NSubstitute, a fake <code>HttpMessageHandler</code>	respx	Fakes HTTP calls in tests
Kestrel	uvicorn	The web server that runs the app
ASP.NET Core minimal APIs	FastAPI	Routing, model binding, DI, OpenAPI
<code>record</code> + <code>DataAnnotations</code>	Pydantic models	Typed data with validation
<code>IOptions<T></code>	pydantic-settings	Typed configuration from the environment
EF Core	SQLAlchemy 2.0	ORM and query builder
EF Core migrations	Alembic	Schema migrations
<code>HttpClient</code>	httpx	HTTP client, sync and async
<code>Task</code> , <code>async / await</code>	asyncio coroutines	Asynchronous code

The table is a starting point, not a promise. The rest of this chapter covers the four places where the mapping breaks hardest.

0.3 Break 1 — Types are checked, but not enforced

In C#, a type is a contract the compiler enforces and the runtime relies on. In Python, a type hint is an *annotation*: the interpreter stores it and otherwise ignores it.

```
def greet(name: str) -> str:
    return "Hello, " + name

greet(42)  # Runs. Fails only inside, at the "+", with a TypeError.
```

Nothing stops that call at run time. What stops it is **pyright**, run before the code ships — exactly the role the C# compiler plays, except that it is a separate tool and you could skip it. Lustjinn runs pyright in **strict** mode, in CI, on every push: that is our `TreatWarningsAsErrors`.

THE .NET LENS

`str | None` is Python's `string?`. With nullable reference types on, C# warns when you dereference something that might be null; pyright strict does the same for `None`. The difference is that C#'s `null` checks are warnings you opted into, while in strict pyright they are errors from the start.

WORTH KNOWING

Some libraries *do* read type hints at run time, on purpose. **Pydantic** and **FastAPI** are the two you will meet first: a parameter declared `limit: int` makes FastAPI parse and validate the query string as an integer and return a 422 if it is not. That is model binding, driven by the same annotations pyright checks.

0.4 Break 2 — There is no build step

`dotnet build` turns your project into an assembly; `dotnet run` runs the assembly. Python has no equivalent step you run yourself. The interpreter (**CPython**, the standard one) compiles each file to bytecode the first time it is imported, caches it in a `__pycache__` folder, and executes it. A syntax error surfaces when the file is loaded; a typo in a name surfaces only when that line runs.

That is why the tooling matters more than in .NET: **ruff** and **pyright** are the build. If they are clean and the tests pass, the code is as checked as it will get before it runs.

0.5 Break 3 — Every project gets its own interpreter environment

NuGet keeps packages in a global cache, and each project references the versions it wants; two projects can use different versions of the same package without noticing each other.

Python installs packages *into an interpreter*. Install a package into the system Python and every project on the machine sees that version. The answer is a **virtual environment**: a folder (`.venv`) holding a private copy of the interpreter's package directory, one per project.

uv manages all of it. It downloads the Python version the project pins, creates `.venv`, installs exactly what `uv.lock` says, and runs commands inside that environment:

<code>uv sync</code>	<code># make .venv match uv.lock</code>	<code>≈ dotnet restore</code>
<code>uv add httpx</code>	<code># add a dependency, update the lock</code>	<code>≈ dotnet add package</code>
<code>uv run pytest</code>	<code># run a command inside .venv</code>	<code>≈ dotnet test</code>
<code>uv python install</code>	<code># fetch the pinned Python version</code>	<code>≈ installing an SDK</code>

WATCH OUT

Your Windows machine has an older system Python (3.9) on the `PATH`. Lustjinn never uses it: `uv` fetches the version `.python-version` asks for and keeps it separate. If a command ever behaves as if it were on 3.9, it was run with `python` instead of `uv run python`.

0.6 Break 4 — Async runs on one thread

In ASP.NET Core, each request runs on a thread-pool thread. If a handler blocks — a synchronous database call, a `Thread.Sleep` — it wastes that one thread, and the pool covers for it.

Python's `asyncio` runs every coroutine on a **single thread**, inside an **event loop**. An `await` hands control back to the loop so another request can make progress. A call that blocks without awaiting does not waste one thread; it freezes *every* request the server is handling until it returns.

```
async def handler():
    time.sleep(2)           # Blocks the event loop: the whole server stops for 2 s.
    await asyncio.sleep(2)  # Yields to the loop: other requests keep running.
```

This is why Lustjinn uses the async versions of everything that waits: **httpx**'s `AsyncClient`, SQLAlchemy's async engine, an async Postgres driver.

THE .NET LENS

`async def` $\approx$ an `async Task` method, and `await` means the same thing. The difference is what happens *between* awaits: in .NET, other work runs on other threads; in Python, other work runs only when you await. There is no `ConfigureAwait`, and no `SynchronizationContext` to think about.

WORTH KNOWING

FastAPI hides one sharp edge for you. An endpoint declared with plain `def` (not `async def`) is run on a separate thread pool, so blocking inside it is safe. An endpoint declared `async def` runs on the event loop and must never block.

0.7 Smaller differences you will meet soon

- **Files are modules; folders are packages.** There is no `namespace` keyword and no project reference: `from lustjinn.stories import Story` imports by path from the package folder.
- **Naming:** `snake_case` for functions, variables and modules; `PascalCase` for classes; `UPPER_CASE` for constants. The leading underscore (`_helper`) means "private by convention" — nothing enforces it.
- **self is explicit.** A method's first parameter is the instance, written out. The constructor is `__init__`.
- **Indentation is syntax.** Blocks have no braces; the indentation is the block. `ruff`'s formatter keeps it consistent.
- `None`, `True`, `False` are capitalised.

0.8 What is already on your machine

The setup for this course and for step 1 is installed:

- **uv** — the Python toolchain. It will fetch the current Python on first use.
- **pandoc** and **Typst** — they turn the Markdown chapters of this course into this PDF.

The course lives in the repository under `docs/course/`. To rebuild it:

```
./docs/course/build.ps1          # the dark edition
./docs/course/build.ps1 -Light    # a light edition, for printing
```

TRY IT

Run `uv --version` and `uv python list`. The second command lists the Python versions uv can install, and any it already has. Next chapter, one of them becomes Lustjinn's.

0.9 What is next

Chapter 1 — Hello, deployed. A uv project, FastAPI with one endpoint, one pytest test, ruff and pyright in CI, and the whole thing deployed to Render — so that `GET /health` answers from the internet. It teaches the project layout, uv in practice, and how a Python web app is actually served.

CHAPTER 1

Hello, deployed

The goal of step 1 is the smallest thing that is real: a Python web app with one endpoint, one test, the quality gates, and a public URL that answers. Nothing about roleplay yet. Everything later grows out of this skeleton, so it is worth understanding every file in it.

The step lands in three pieces, and this chapter grows with them:

1. **The project** — uv, FastAPI, one endpoint, one test, ruff and pyright.
2. **CI** — the gates running on Forgejo on every push.
3. **The deploy** — Render serving the app from the GitHub mirror. *(Coming with issue #5.)*

1.1 What uv created

One command started the project:

```
uv init --package --name lustjinn --python 3.14
```

and two more added the libraries:

```
uv add fastapi "uvicorn[standard]"
uv add --dev pytest httpx2 ruff pyright
```

The result, ignoring the docs:

<code>.python-version</code>	3.14 – which Python this project runs on
<code>pyproject.toml</code>	the project file
<code>uv.lock</code>	every package, pinned to an exact version and hash
<code>.venv/</code>	the project's private environment (gitignored)
<code>src/lustjinn/</code>	
<code>__init__.py</code>	makes the folder a package
<code>main.py</code>	the FastAPI app
<code>tests/</code>	
<code>test_health.py</code>	the one test

Python 3.14 is pinned because it is the newest stable release: 3.15 exists, but only as a release candidate. uv downloaded 3.14.7 on its own, without touching the Python 3.9 already installed on the machine.

1.1.1 Why the code lives under `src/`

The package is `src/lustjinn/`, not `lustjinn/` at the root. With the code at the root, `import lustjinn` would work in tests simply because the current folder happens to be on the import

path — and a test could pass against files that would never be part of the installed app. Under `src/`, the only way to import `lustjinn` is to install it, which `uv` does: `uv sync` installs the project into `.venv` in *editable* mode, meaning the environment points at `src/` instead of copying it, so edits are live.

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This is the test project's `<ProjectReference>`. A test project in .NET cannot see the app's source files, only the assembly the reference builds. The `src/` layout gives Python the same discipline: tests see the package as installed, the way the server will.

1.2 The project file

`pyproject.toml` is the `.csproj`. It is TOML, a format of `[tables]` and `key = value` lines.

```
[project]
name = "lustjinn"
version = "0.1.0"
requires-python = ">=3.14"
dependencies = [
    "fastapi>=0.142.2",
    "uvicorn[standard]>=0.54.0",
]

[dependency-groups]
dev = [
    "httpx2>=2.13.1",
    "pyright>=1.1.414",
    "pytest>=9.1.1",
    "ruff>=0.16.10",
]
```

- **[project]** is the standard part every Python tool understands: name, version, the Python it needs, and the packages the app needs at run time.
- **uvicorn[standard]** — the brackets name an *extra*: optional dependencies a package offers. `standard` adds uvicorn's faster event loop and HTTP parser, and file watching for `--reload`.
- **[dependency-groups]** holds what development needs but production does not: the test runner and the gates. A production install leaves them out.
- **[build-system]** says how to turn the project into an installable package (`uv_build`).
- **[tool...]** tables configure each tool — `[tool.ruff]`, `[tool.pyright]`, `[tool.pytest...]` — all in the one file.

THE .NET LENS

`"fastapi>=0.142.2"` reads like NuGet's `Version="0.142.2"`, and both mean *at least* that version. The difference is which version gets picked: NuGet resolves the **lowest** version that satisfies every constraint, `uv` the **highest**. Either way, the lock file is what makes the build repeatable — `uv.lock` plays the role of `packages.lock.json`, and it is committed.

1.3 The app

The whole application, `src/lustjinn/main.py`:

```
"""The FastAPI application: the object uvicorn serves."""

from fastapi import FastAPI

app = FastAPI(title="Lustjinn")

@app.get("/health")
async def health() -> dict[str, str]:
    """Answers while the process is up. Render and the PWA's waking screen poll it."""
    return {"status": "ok"}
```

Line by line:

- The string at the top of a file, function or class is a **docstring**: documentation the language keeps (`help()` shows it). It plays the part of an XML `<summary>` comment.
- `from fastapi import FastAPI` imports one name from a package — a `using` that names exactly what it brings in.
- `app = FastAPI(...)` creates the application at module level. When Python imports this file, the line runs, and `app` exists from then on.
- `@app.get("/health")` is a **decorator**. Python calls `app.get("/health")`, which returns a function; that function receives `health`, registers it as the handler for `GET /health`, and hands it back unchanged. The `@` line is shorthand for exactly that.
- `async def health() -> dict[str, str]` declares a coroutine that returns a dictionary of strings. FastAPI reads the return annotation: it documents the response in the OpenAPI schema and validates what the function returns.
- `return {"status": "ok"}` — FastAPI serialises the dictionary to JSON.

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Put side by side with a minimal API, there is almost nothing to translate:

```
var app = WebApplication.CreateBuilder(args).Build();
app.MapGet("/health", () => new Dictionary<string, string> { ["status"] = "ok" });
app.Run();
```

`app.MapGet(...)` becomes the decorator, and `Dictionary<string, string>` becomes `dict[str, str]`. What has no counterpart is `app.Run()` — see the next section.

1.4 How Python is served

A .NET app is an executable: `Program.cs` builds the app and `app.Run()` starts Kestrel inside the same process. A Python web app is not run; it is **imported by a server**. The server here is **uvicorn**:

```
uv run uvicorn lustjinn.main:app --reload
```

`lustjinn.main:app` means *import the module `lustjinn.main` and serve the object called `app` in it*. `uvicorn` owns the process, the socket and the event loop; FastAPI only answers requests handed to it. `--reload` restarts the server when a file changes — for development only.

The contract between the two is **ASGI** (Asynchronous Server Gateway Interface): a standard signature every async Python server can call and every async framework implements. That is why FastAPI could run on another ASGI server without changing a line.

THE .NET LENS

`uvicorn` is Kestrel, and ASGI is the boundary between Kestrel and the ASP.NET Core pipeline — except that in Python the boundary is a public standard, so servers and frameworks mix and match.

With the server running, FastAPI also serves its own documentation: `/docs` is an interactive Swagger UI and `/openapi.json` the schema, both generated from the code — no Swashbuckle to add.

1.5 The test

`tests/test_health.py`:

```
from fastapi.testclient import TestClient

from lustjinn.main import app

client = TestClient(app)

def test_health_answers_ok() -> None:
    response = client.get("/health")

    assert response.status_code == 200
    assert response.json() == {"status": "ok"}
```

- **pytest finds tests by name**: files called `test_*.py`, functions called `test_*`. No attribute, no test class needed.
- **Plain `assert`**: `pytest` rewrites `assert` statements when it loads a test file, so a failure shows both sides of the comparison — no `Assert.Equal` family to learn.
- **`TestClient`** runs the app in-process and speaks to it over ASGI directly: no server, no port, no network.

THE .NET LENS

`TestClient(app)` is `WebApplicationFactory<Program>` plus `CreateClient()`, in one line.

WATCH OUT

When this test was first written, the dev dependency was `httpx`, as countless tutorials say. The test passed — with a deprecation warning, because `Starlette` (the toolkit under FastAPI) now wants its successor, `httpx2`. `pyright` went further and failed: with `httpx2` missing, it could not know the types of `client.get`, `response.status_code` or `response.json`, and strict mode refuses unknown types. Two lessons in one. Tutorials age fast in Python — check what the

library itself asks for. And strict typing earns its keep on day one: it turned a quiet warning into an error.

1.6 The gates

Four commands decide whether a change is acceptable. All four are clean on this commit:

```
uv run pytest          # the tests
uv run ruff check       # lint
uv run ruff format --check # formatting, without rewriting
uv run pyright          # types, strict
```

ruff is configured in `pyproject.toml` with a deliberate set of rule families. Most are style and likely-bug checks. One deserves a mention: **ASYNC** flags blocking calls made inside `async def` functions — the event-loop freeze from chapter 0, caught before it runs.

pyright runs in `strict` mode over `src` and `tests`. Strict means every value must have a type pyright can know; an unknown type is an error, not a shrug.

TRY IT

Start the server with `uv run uvicorn lustjinn.main:app --reload` and open `http://127.0.0.1:8000/docs`. Call `/health` from the Swagger page.

Then, in `main.py`, change the return to `{"status": 1}` and run `uv run pyright`. The annotation says `dict[str, str]`; pyright says no. Change it back.

1.7 CI on Forgejo

The gates are only worth something if nothing reaches `develop` without passing them. A workflow in `.forgejo/workflows/ci.yml` runs all four on Forgejo, the self-hosted forge that is `origin`:

```
on:
  push:
    branches: [main, develop]
  pull_request:
    branches: [main, develop]

jobs:
  gates:
    runs-on: ubuntu-latest
    env:
      UV_VERSION: "0.12.21"
    steps:
      - uses: actions/checkout@v5
        with:
          persist-credentials: false
      - name: Install uv
        run: |
          curl -LsSf "https://astral.sh/uv/${UV_VERSION}/install.sh" | sh
          echo "$HOME/.local/bin" >> "$GITHUB_PATH"
      - name: Sync the environment from uv.lock
        run: uv sync --locked
```

```

- name: ruff check
  run: uv run ruff check
- name: ruff format
  run: uv run ruff format --check
- name: pyright (strict)
  run: uv run pyright
- name: pytest
  run: uv run pytest

```

The syntax is GitHub Actions'; Forgejo runs the same files. Reading it top to bottom:

- **on** — a pull request into `develop` or `main` runs the gates, and so does the push that lands when it merges. Pushes to a feature branch alone do not: the pull request covers them, without running everything twice.
- **runs-on: ubuntu-latest** — on this forge, the name of a Linux container image on the runner. Each run starts from a fresh container, and `uv` and the project's Python come from the steps below — so a green run proves the project builds from its committed files alone.
- **Install uv** — one pinned release, the same version as on the development machine. Python then comes from `uv`, exactly as on Windows: `uv sync` reads `.python-version` and fetches 3.14.
- **uv sync --locked** — creates the environment from `uv.lock`, and **fails** if the lock no longer matches `pyproject.toml`. Without `--locked`, `uv` would quietly re-resolve, and CI would test versions nobody committed.
- **The gates, cheapest first** — a formatting slip fails in seconds, before `pyright` and the tests run.

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`uv sync --locked` is `dotnet restore --locked-mode:restore` exactly what the lock file says, or fail. And the workflow as a whole is the YAML pipeline you would write for Azure DevOps or GitHub Actions around `dotnet build` and `dotnet test` — with `uv sync` as the restore and `pyright` and `ruff` standing in for the build.

WORTH KNOWING

The workflow lives in `.forgejo/`, not `.github/`. GitHub ignores `.forgejo/`, so the GitHub mirror — which only exists for Render to deploy from — never runs it and never spends Actions minutes. The other house conventions (the `bash -e` shell, a checkout that keeps no credentials) are shared with the owner's other repositories.

1.8 Still to come in this chapter

- **The deploy** — Render serving the app, and `GET /health` answering from the internet (issue #5).